



Test and Reliability

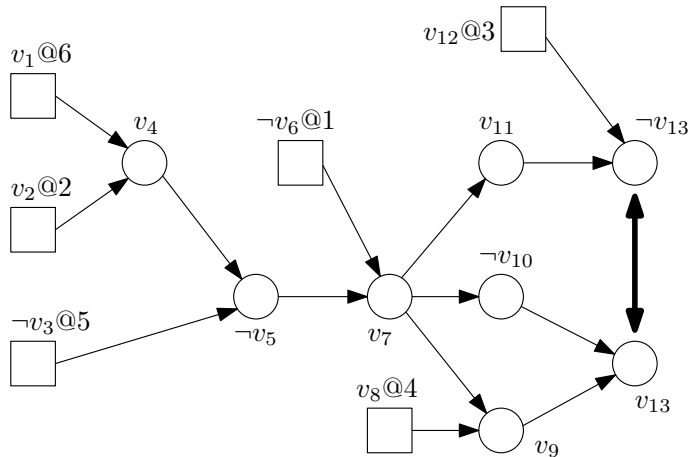
Exercise Sheet 5

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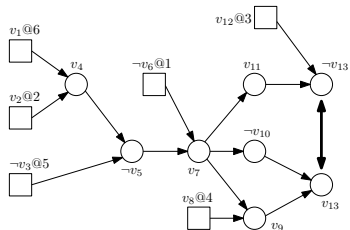
July 9, 2026



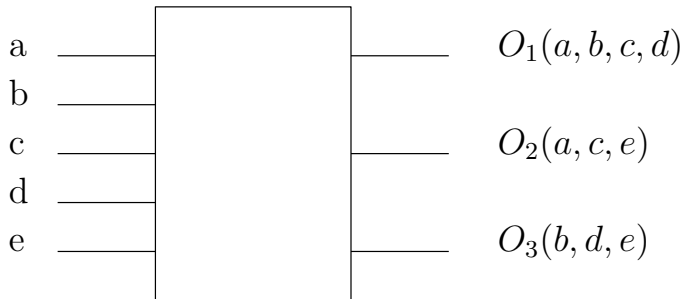
Exercise 1) First Unique Implication Point (1UIP)



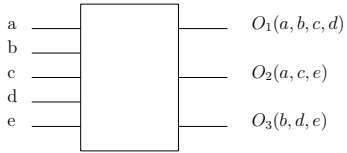
Exercise 1) First Unique Implication Point (1UIP)



Exercise 2) Pseudo-exhaustive Testing



Exercise 2a) Pseudo-exhaustive Testing



	<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>
O_1					
O_2					
O_3					

Exercise 2b) Pseudo-exhaustive Testing

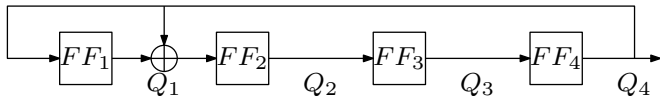
Minimal example with $I = O = 3$, $w = 2$, $p = 3$. Obviously, $w < p$.

	a	b	c
O_1	1	1	0
O_2	1	0	1
O_3	0	1	1

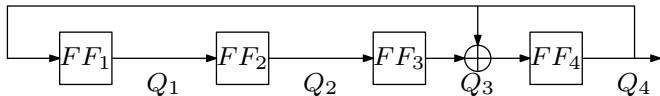
To do: Is it really minimal? Proof:

Exercise 3) Linear Feedback Shift Registers

Option 1:



Option 2:



Exercise 3) Linear Feedback Shift Registers

LFSR 1				LFSR 2			
Q_1	Q_2	Q_3	Q_4	Q_1	Q_2	Q_3	Q_4
0	0	0	1	0	0	0	1
1	1	0	0	1	0	0	1
0	1	1	0	1	1	0	1
0	0	1	1	1	1	1	1
1	1	0	1	1	1	1	0
1	0	1	0	0	1	1	1
0	1	0	1	1	0	1	0
1	1	1	0	0	1	0	1
0	1	1	1	1	0	1	1
1	1	1	1	1	1	0	0
1	0	1	1	0	1	1	0
1	0	0	1	0	0	1	1
1	0	0	0	1	0	0	0
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	1	0	0	0	1

Exercise 4) Polynomials

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